

Do **not** write solutions on this page.

12. [Maximum mark: 20]

Consider $\phi = (a + bi)^3$, where $a, b \in \mathbb{R}$.

(a) In terms of a and b , find

(i) the real part of ϕ ;

(ii) the imaginary part of ϕ .

[3]

(b) Hence, or otherwise, show that $(1 + \sqrt{3}i)^3 = -8$.

[2]

The roots of the equation $z^3 = -8$ are u , v and w , where $u = 1 + \sqrt{3}i$ and $v \in \mathbb{R}$.

(c) Write down v and w , giving your answers in Cartesian form.

[2]

On an Argand diagram, u , v and w are represented by the points U , V and W respectively.

(d) Find the area of the triangle UVW .

[3]

Each of the points U , V and W is rotated counter-clockwise (anticlockwise) about 0 through an angle of $\frac{\pi}{4}$ to form three new points U' , V' and W' . These points represent the complex numbers u' , v' and w' respectively.

(e) Find u' , v' and w' , giving your answers in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$.

[4]

(f) Given that u' , v' and w' are the solutions of $z^3 = c + di$, where $c, d \in \mathbb{R}$, find the value of c and the value of d .

[3]

It is given that u , v , w , u' , v' and w' are all solutions of $z^n = \alpha$ for some $\alpha \in \mathbb{C}$, where $n \in \mathbb{N}$.

(g) Find the smallest positive value of n .

[3]

